Quant_day_2

March 5, 2026

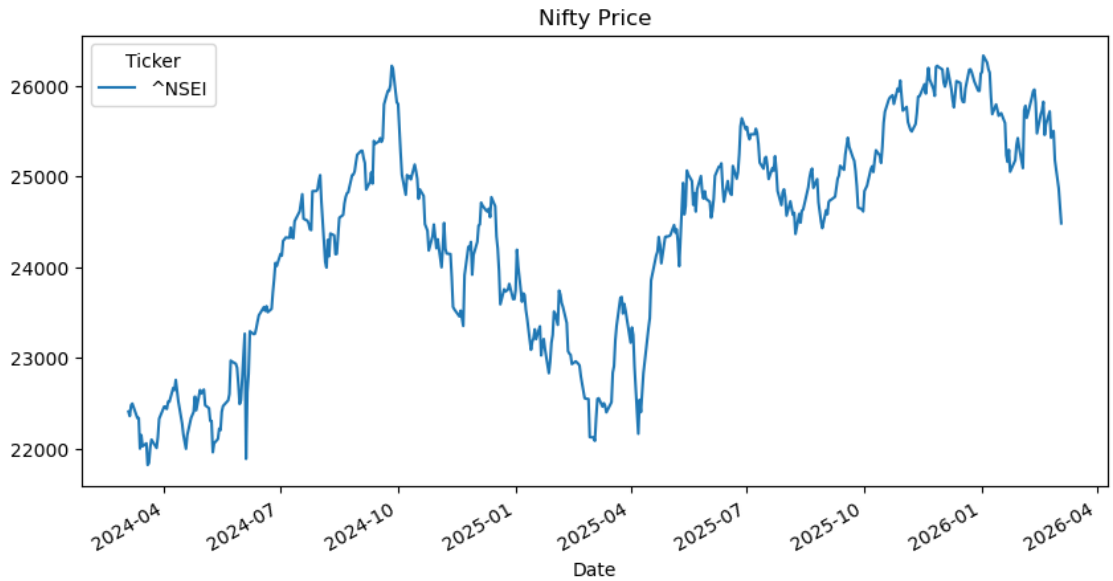
[1]:
```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt
```

[2]:
```
data = yf.download("^NSEI", period="2y")

price = data["Close"]
price.head()
```

```
C:\Users\MD RAJA\AppData\Local\Temp\ipykernel_10956\1720156814.py:1:
FutureWarning: YF.download() has changed argument auto_adjust default to True
  data = yf.download("^NSEI", period="2y")
[*********************100%***********************]  1 of 1 completed
```

[2]:
```
Ticker             ^NSEI
Date
2024-03-04  22405.599609
2024-03-05  22356.300781
2024-03-06  22474.050781
2024-03-07  22493.550781
2024-03-11  22332.650391
```

[3]:
```
price.plot(figsize=(10,5))
plt.title("Nifty Price")
plt.show()
```



[5]:
```
from statsmodels.tsa.stattools import adfuller
adf_price = adfuller(price.dropna())

print("ADF Statistic:", adf_price[0])
print("p-value:", adf_price[1])
```

```
ADF Statistic: -2.215787882608344
p-value: 0.20056534647536545
```

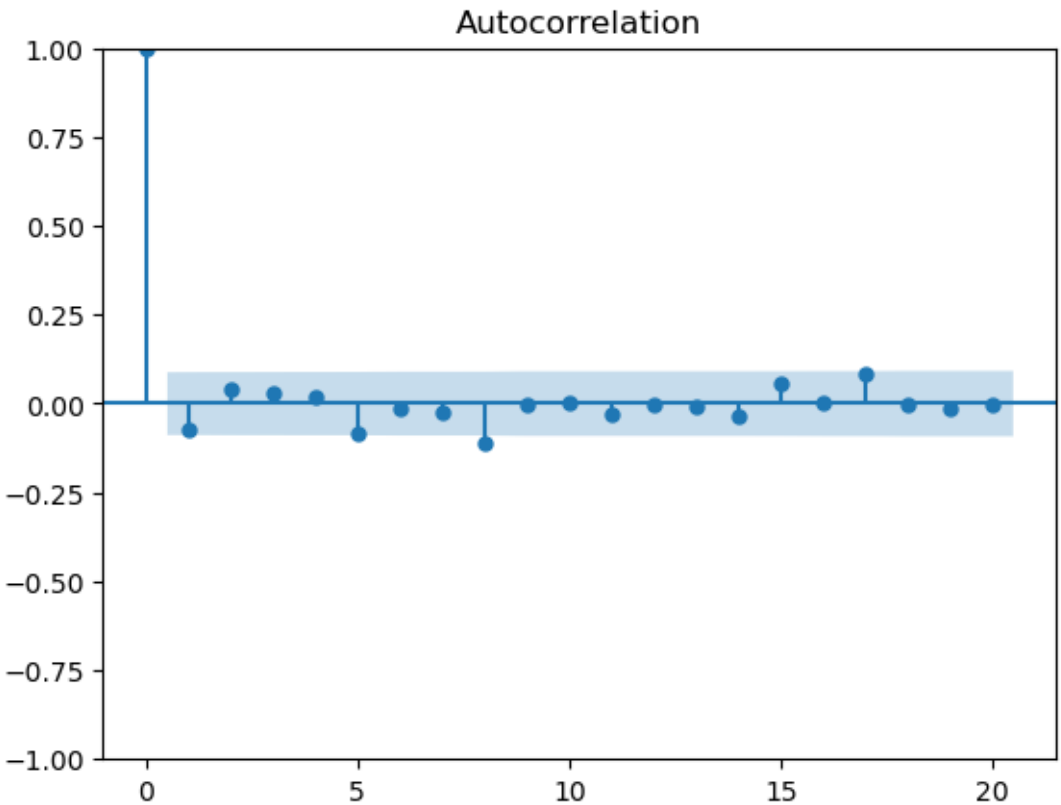
since p -value adf test is greater than 0.05. so price is non-stationary

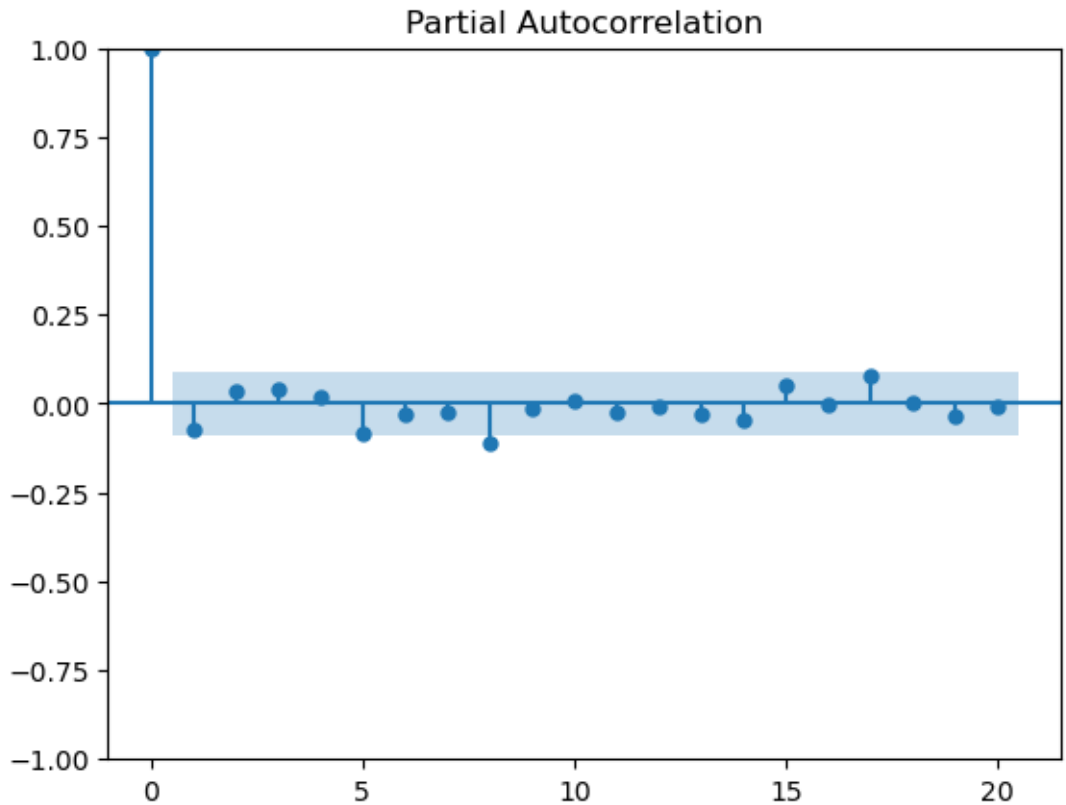
[6]:
```
returns = np.log(price / price.shift(1))
returns = returns.dropna()

returns.plot(figsize=(10,5))
plt.title("Log Returns")
plt.show()
```

[7]:
```
adf_returns = adfuller(returns)

print("ADF Statistic:", adf_returns[0])
print("p-value:", adf_returns[1])
```

```
ADF Statistic: -23.76232778473388
p-value: 0.0
```

[9]:
```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
plot_acf(returns, lags=20)
plt.show()

plot_pacf(returns, lags=20)
plt.show()
```


Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20

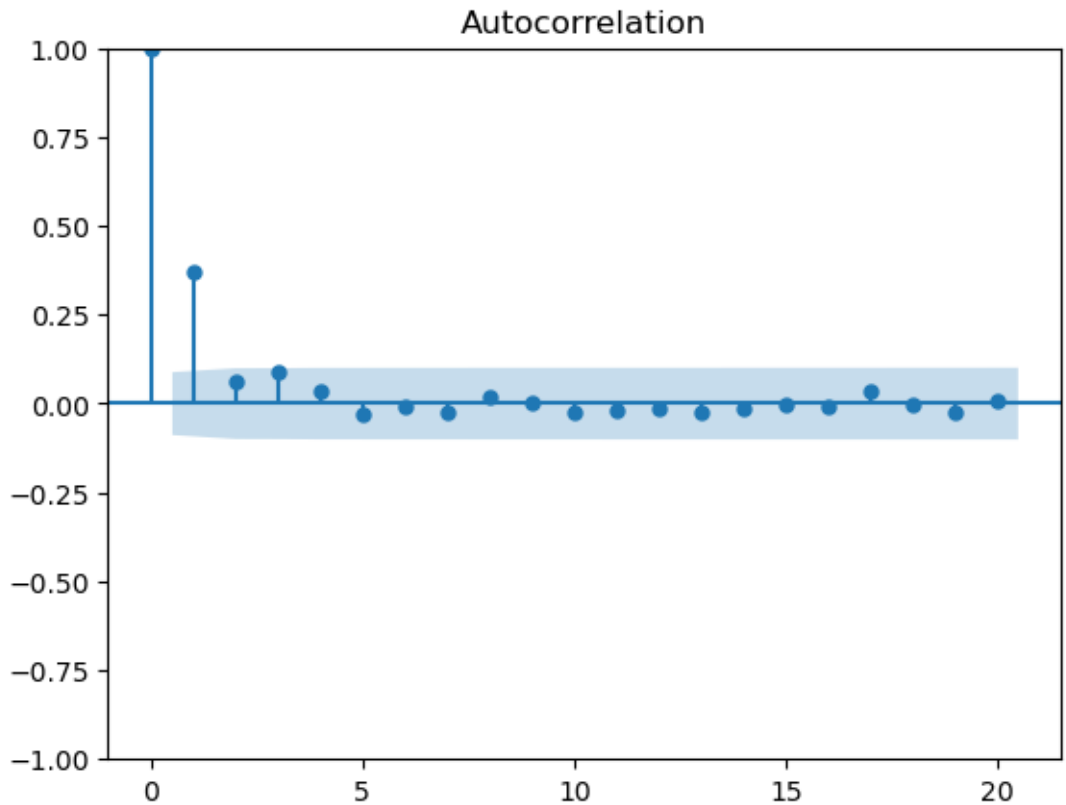


[11]:
```
from statsmodels.stats.diagnostic import acorr_ljungbox
lb = acorr_ljungbox(returns, lags=[10], return_df=True)
print(lb)
```

```
      lb_stat  lb_pvalue
10  14.751792   0.141376
```

p>0.05 no autocorrelation.

[12]:
```
plot_acf(returns**2, lags=20)
plt.show()
```



[27]:
```
from arch import arch_model
garch = arch_model(returns, vol='Garch', p=1, q=1)

result = garch.fit()

print(result.summary())
```

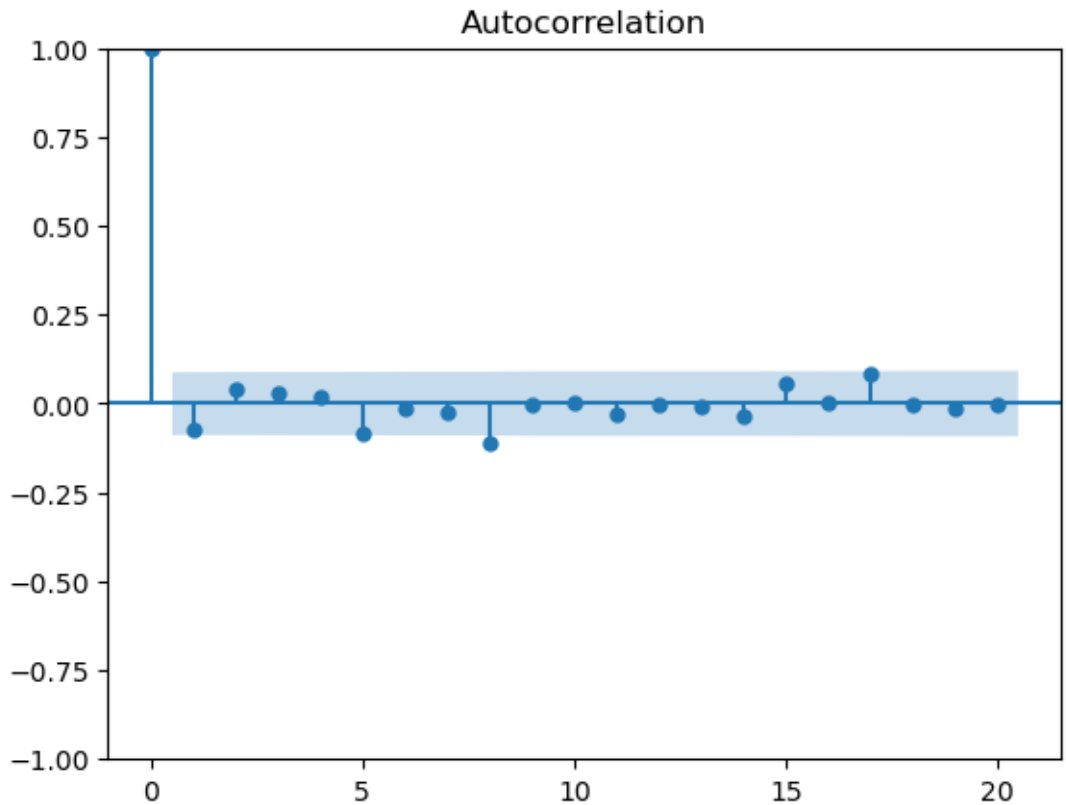
```
Iteration:      1,   Func. Count:      6,   Neg. LLF: 2059647957.2151713
Iteration:      2,   Func. Count:     18,   Neg. LLF: 21794402.167837776
Optimization terminated successfully    (Exit mode 0)
            Current function value: -1705.4735808896353
            Iterations: 2
            Function evaluations: 27
            Gradient evaluations: 2
                     Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:                   ^NSEI   R-squared:                       0.000
Mean Model:             Constant Mean   Adj. R-squared:                  0.000
Vol Model:                      GARCH   Log-Likelihood:                1705.47
Distribution:                  Normal   AIC:                          -3402.95
Method:            Maximum Likelihood   BIC:                          -3386.15
```

```
                                      No. Observations:                  493
Date:                Thu, Mar 05 2026   Df Residuals:                      492
Time:                        00:47:32   Df Model:                            1
                                 Mean Model
==============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
mu         3.8128e-04  3.630e-04      1.050      0.294 [-3.302e-04,1.093e-03]
                            Volatility Model
===============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
-------------------------------------------------------------------------------
omega      3.3501e-05  8.263e-07     40.546      0.000 [3.188e-05,3.512e-05]
alpha[1]       0.2000  9.573e-02      2.089  3.668e-02   [1.238e-02,  0.388]
beta[1]        0.3000  6.712e-02      4.470  7.838e-06     [  0.168,  0.432]
===============================================================================

Covariance estimator: robust
```

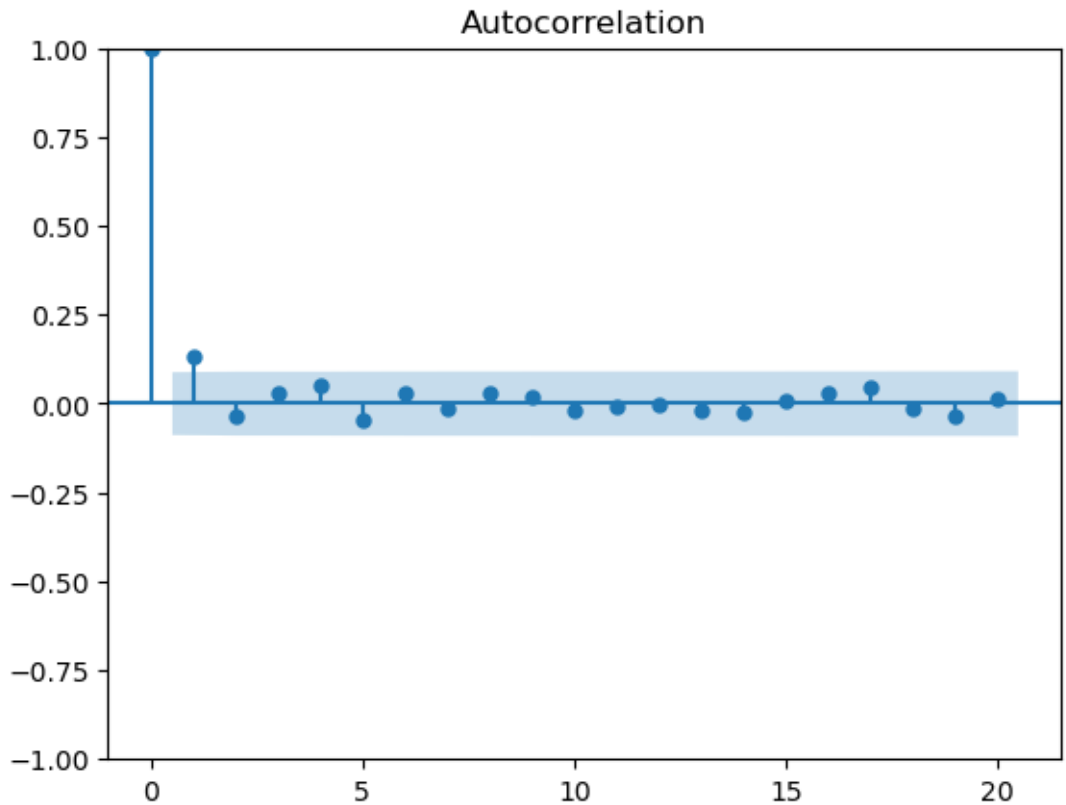
```
C:\Users\MD RAJA\AppData\Roaming\Python\Python313\site-
packages\arch\univariate\base.py:694: DataScaleWarning: y is poorly scaled,
which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 6.7e-05. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 100 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
```

[30]:
```
sigma = result.conditional_volatility
```

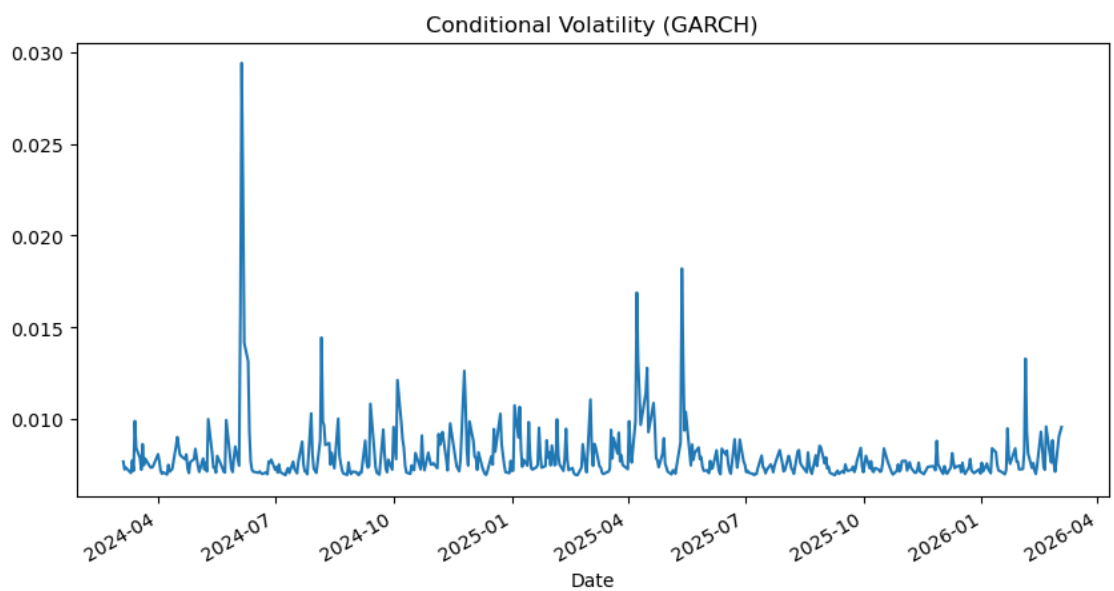
[31]:
```
plot_acf(result.resid, lags=20)
plt.show()
```



[32]:
```
plot_acf(result.std_resid**2, lags=20)
plt.show()
```



[33]:
```
result.conditional_volatility.plot(figsize=(10,5))
plt.title("Conditional Volatility (GARCH)")
plt.show()
```



[34]:
```
forecast = result.forecast(horizon=5)

print(forecast.variance.tail())
```

```
                 h.1      h.2       h.3       h.4      h.5
Date
2026-03-04  0.000112  0.00009  0.000078  0.000073  0.00007
```

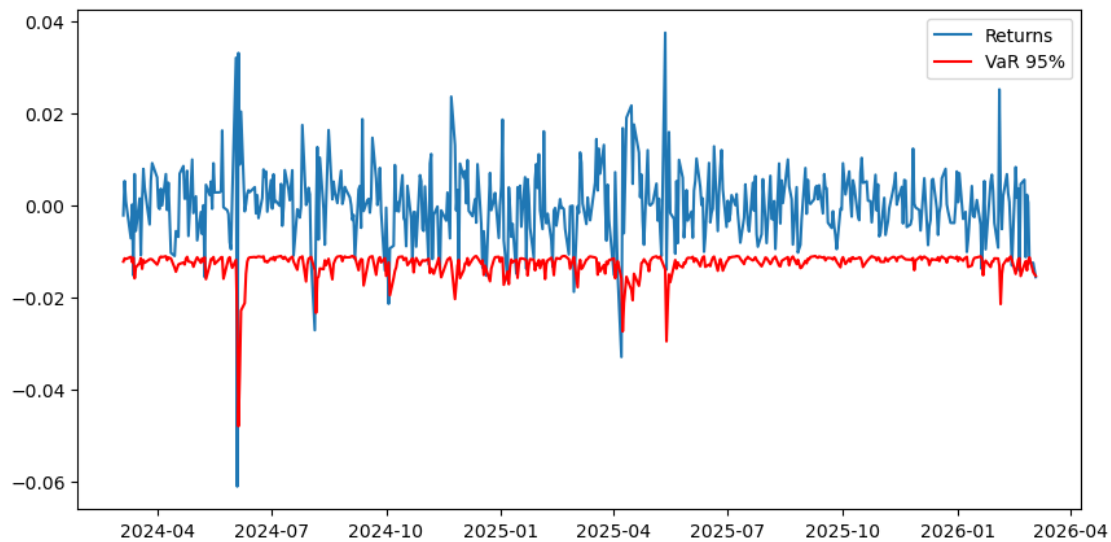
[35]:
```
variance = forecast.variance.iloc[-1,0]
volatility = np.sqrt(variance)

mu = result.params['mu']

z = -1.645   # 95% confidence

VaR = mu + z * volatility

print("1-day 95% VaR:", VaR)
```

```
1-day 95% VaR: -0.017033455706563717
```

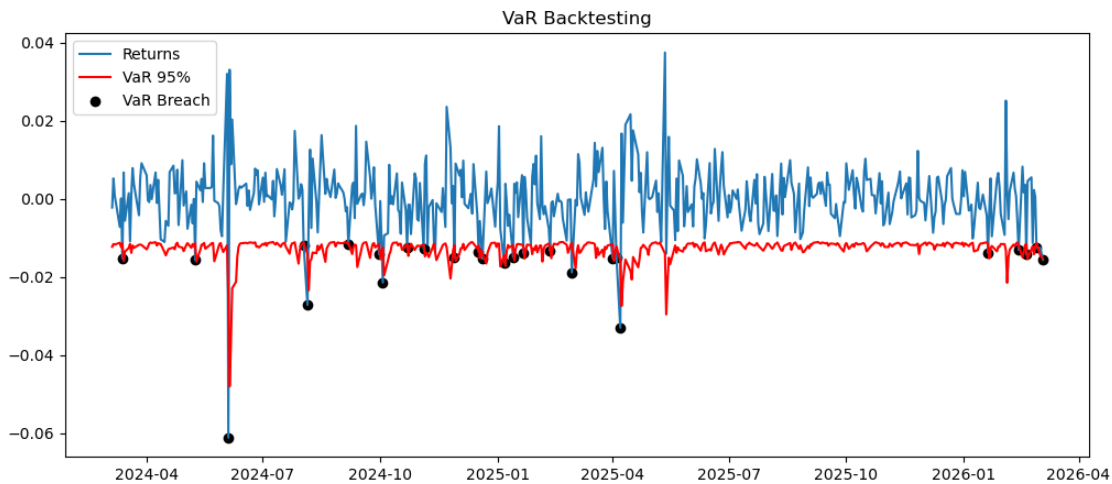
[39]:
```
# conditional volatility
sigma = result.conditional_volatility

# mean
mu = result.params['mu']

# VaR
z = -1.645
VaR_series = mu + z * sigma

# make returns a series
returns = returns.squeeze()

# align
returns, VaR_series = returns.align(VaR_series)

# violations
violations = returns < VaR_series
violation_rate = violations.mean()

print("Violation Rate:", violation_rate)
```

```
Violation Rate: 0.05273833671399594
```

[40]:
```
plt.figure(figsize=(10,5))
plt.plot(returns, label="Returns")
plt.plot(VaR_series, color="red", label="VaR 95%")
plt.legend()
plt.show()
```



[41]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(12,5))
plt.plot(returns, label="Returns")
plt.plot(VaR_series, color="red", label="VaR 95%")
plt.scatter(returns.index[violations], returns[violations],
            color="black", label="VaR Breach")
plt.legend()
plt.title("VaR Backtesting")
plt.show()
```



[42]:
```
import numpy as np

n = len(returns)
x = violations.sum()
p = 0.05

LR = -2 * np.log(((1-p)**(n-x) * p**x) /
                 ((1-x/n)**(n-x) * (x/n)**x))

print("Kupiec Test Statistic:", LR)
```

```
Kupiec Test Statistic: 0.0765162165544527
```

[43]:
```
garch_t = arch_model(returns, vol='Garch', p=1, q=1, dist='t')
res_t = garch_t.fit()
print(res_t.summary())
```

```
Iteration:      1,   Func. Count:      7,   Neg. LLF: 15400.662595942198
Iteration:      2,   Func. Count:     23,   Neg. LLF: 63165.72408775753
Iteration:      3,   Func. Count:     39,   Neg. LLF: 5834.726284559738
Iteration:      4,   Func. Count:     46,   Neg. LLF: 5860.402641649316
Iteration:      5,   Func. Count:     53,   Neg. LLF: 5225.68950986396
Iteration:      6,   Func. Count:     60,   Neg. LLF: 3087.6522641317956
Iteration:      7,   Func. Count:     66,   Neg. LLF: 2065529.087742481
Iteration:      8,   Func. Count:     79,   Neg. LLF: 1714.3547035580978
Iteration:      9,   Func. Count:     85,   Neg. LLF: -985.133246482855
Iteration:     10,   Func. Count:     90,   Neg. LLF: -985.1332465117091
Inequality constraints incompatible    (Exit mode 4)
            Current function value: -985.133246482855
            Iterations: 11
            Function evaluations: 90
```

```
            Gradient evaluations: 10
                        Constant Mean - GARCH Model Results
==============================================================================
====
Dep. Variable:                        ^NSEI   R-squared:
0.000
Mean Model:                    Constant Mean   Adj. R-squared:
0.000
Vol Model:                             GARCH   Log-Likelihood:
985.133
Distribution:      Standardized Student's t   AIC:
-1960.27
Method:                  Maximum Likelihood   BIC:
-1939.26
                                              No. Observations:
493
Date:                      Thu, Mar 05 2026   Df Residuals:
492
Time:                              00:55:34   Df Model:
1
                                 Mean Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
mu             0.0206  2.240e-03      9.183  4.209e-20 [1.618e-02,2.495e-02]
                              Volatility Model
==============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
omega      6.7002e-13  4.188e-04  1.600e-09      1.000 [-8.209e-04,8.209e-04]
alpha[1]       1.0000      5.594      0.179      0.858      [ -9.965, 11.965]
beta[1]        0.0000  2.291e-03      0.000      1.000 [-4.491e-03,4.491e-03]
                                 Distribution
==============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
nu             6.1742    176.962  3.489e-02      0.972 [-3.407e+02,3.530e+02]
==============================================================================

Covariance estimator: robust
WARNING: The optimizer did not indicate successful convergence. The message was
Inequality constraints incompatible.
See convergence_flag.

C:\Users\MD RAJA\AppData\Roaming\Python\Python313\site-
packages\arch\univariate\base.py:694: DataScaleWarning: y is poorly scaled,
which may affect convergence of the optimizer when
```

```
estimating the model parameters. The scale of y is 6.7e-05. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 100 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
C:\Users\MD RAJA\AppData\Local\Temp\ipykernel_10956\3264403663.py:2:
ConvergenceWarning: The optimizer returned code 4. The message is:
Inequality constraints incompatible
See scipy.optimize.fmin_slsqp for code meaning.

  res_t = garch_t.fit()
```

[44]:
```
forecast = result.forecast(horizon=10)
future_vol = np.sqrt(forecast.variance.iloc[-1])

print("Next 10-day volatility forecast:")
print(future_vol)
```

```
Next 10-day volatility forecast:
h.01    0.010586
h.02    0.009462
h.03    0.008847
h.04    0.008523
h.05    0.008356
h.06    0.008271
h.07    0.008228
h.08    0.008207
h.09    0.008196
h.10    0.008191
Name: 2026-03-04 00:00:00, dtype: float64
```

[ ]:
```

```